Academic-Period and Intersession Characterization of $PM_{2.5}$ Infiltration at Building-Level Indoor Spaces Using Distributed IAQ Monitoring and Spatial Modeling

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ABSTRACT

This study characterizes room-level fine particulate matter ($PM_{2.5}$) in an academic building and evaluates an hourly multizone indoor-air-quality simulation under occupied academic and reduced-occupancy spring-break conditions. Five indoor monitoring locations and one outdoor reference location were instrumented with distributed sensors. Hourly observations were aligned with predictions from a Rhino 8/Grasshopper model using the ANT plug-in. During the academic period, the mean observed indoor concentration was 5.70 $\mu g/m^3$ (2.49 × 10^{-6} grain/ft^3), compared with 4.52 $\mu g/m^3$ (1.97 × 10^{-6} grain/ft^3) simulated. During spring break, the corresponding means were 4.72 and 3.87 $\mu g/m^3$ (2.06 × 10^{-6} and 1.69 × 10^{-6} grain/ft^3). Observed indoor $PM_{2.5}$ declined by 17.1% during the break, while room-average absolute percent bias remained 33.9%-35.8%. The café consistently had the highest concentrations and the largest underprediction; two low-concentration rooms were overpredicted. Results show that occupancy period and room-specific sources or transport pathways should be represented explicitly when calibrating multizone $PM_{2.5}$ models.

INTRODUCTION

Fine particulate matter ($PM_{2.5}$) is an important indoor environmental health concern because exposure is associated with adverse respiratory and cardiovascular outcomes (California Air Resources Board 2015; World Health Organization 2021). Indoor concentrations are influenced by outdoor particle penetration, ventilation, filtration, deposition, resuspension, and indoor generation from activities such as cooking and occupant movement (Abt et al. 2000; Nazaroff 2004; Chen and Zhao 2011). These processes may vary substantially among rooms within the same building (Li et al. 2025).

Academic buildings are especially heterogeneous because classrooms, conference rooms, offices, and food-service areas differ in occupancy, room volume, heating, ventilation, and air conditioning (HVAC) operation, door-opening patterns, and potential indoor sources. Consequently, a building-average concentration may conceal important spatial and temporal differences. Distributed monitoring can identify these contrasts, whereas multizone simulation can evaluate transport mechanisms and support ventilation or filtration decisions when predictions are compared with measurements.

This study combines distributed sensing with a multizone simulation of the Emerging Technology and Entrepreneurship

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Complex (ETEC) at the University at Albany. The objective is to compare measured and simulated hourly $PM_{2.5}$ across five indoor spaces during an academic week and a spring-break week, quantify period-mean model bias, and identify rooms and processes that should be prioritized during future model calibration.

METHODOLOGY

The evaluation paired distributed indoor–outdoor $PM_{2.5}$ monitoring with an hourly multizone airflow and contaminant-transport simulation of the ETEC building. The following subsections describe the sensor deployment and study periods, followed by the multizone model implementation and the metrics used to compare simulated and observed concentrations.

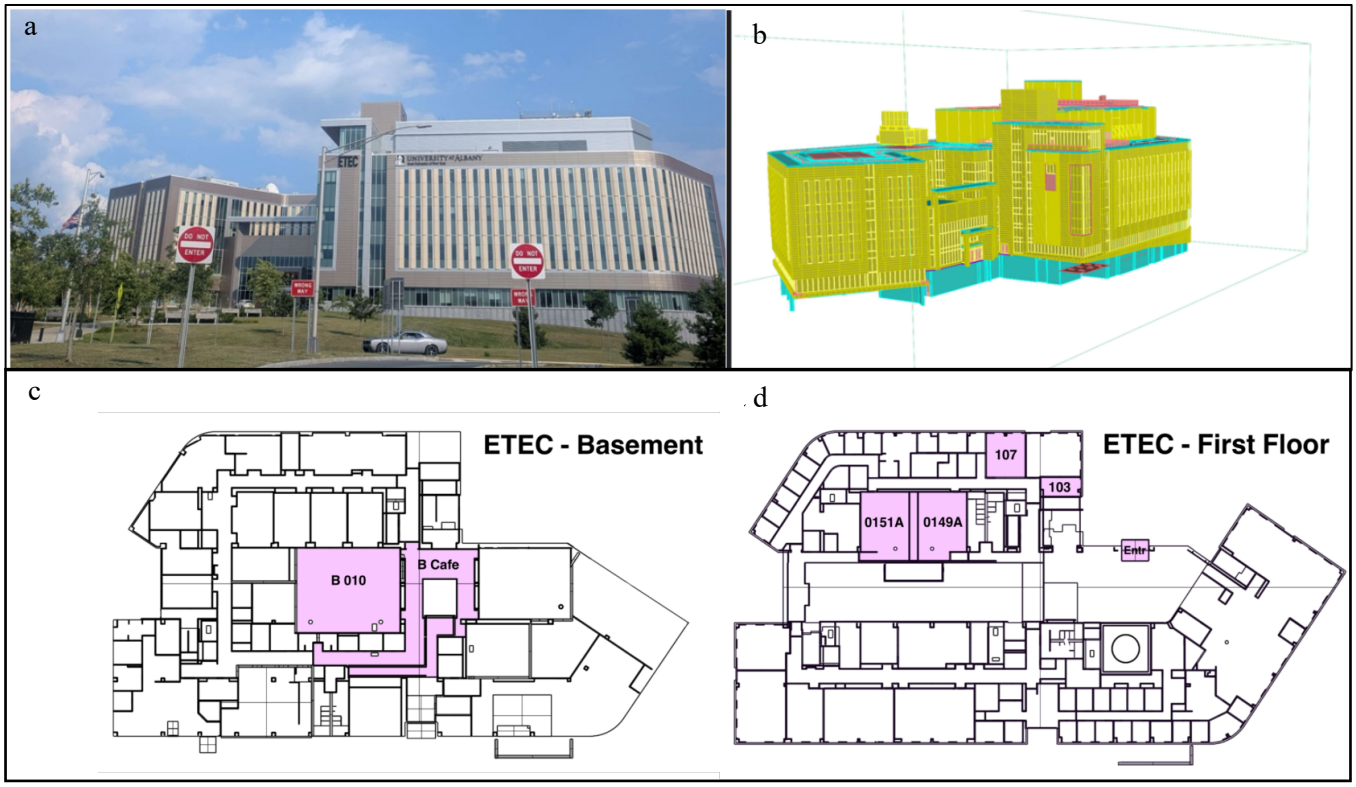


Figure 1 ETEC Building and corresponding floor layouts: (a) exterior view; (b) Rhino 8 wireframe model; (c) basement floor; and (d) first floor. Rooms included in the analysis are highlighted. Rooms 0151A and 0149A form a single large conference room.

Monitoring and Study Periods

Indoor monitors were deployed in ETEC B010, the basement café, ETEC 0149A, ETEC 107, and ETEC 103; a monitor at the ETEC entrance provided an outdoor reference. Sensor data were aggregated to hourly means. The academic evaluation window was March 22 00:00 through March 29 00:00 UTC, 2026, and the spring-break window was March 14 00:00 through March 21 00:00 UTC, 2026, with endpoints included. Missing values were omitted from hourly and period means. Timestamps were harmonized to UTC to resolve the March daylight-saving transition. Because the entrance record ended early and contained substantial gaps, it was used only as contextual outdoor information and was excluded from the indoor-error summaries.

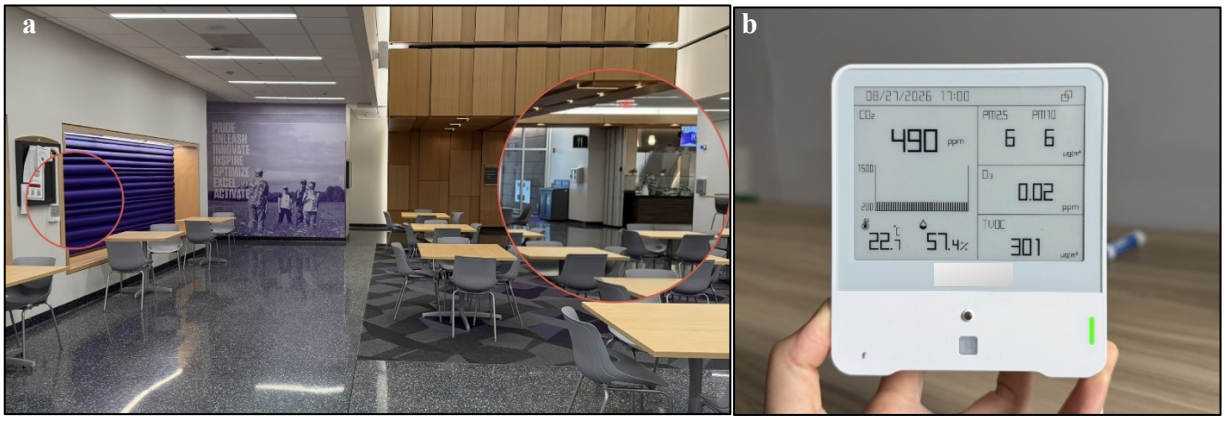


Figure 2 Overview of the ETEC cafe/study area and indoor sensor deployment. (a) Monitored cafe/study area with the sensor location and cafe area highlighted. (b) Close-up of the indoor air-quality sensor.

Multizone Simulation and Evaluation

The building geometry and airflow/contaminant network were implemented in Rhino 8 and Grasshopper with ANT version 0.2.7, which links algorithm-aided building models to multizone indoor-air-quality analysis (Dols and Polidoro 2015; J. Shen et al. 2024). Hourly predicted concentrations were joined to observations by corrected timestamp and room. Evaluation used period means and absolute percent bias, |simulated − observed|/observed × 100%. Indoor summary concentrations are equal-weight means of the five room means; reported summary bias is the mean of room-specific absolute percent biases.

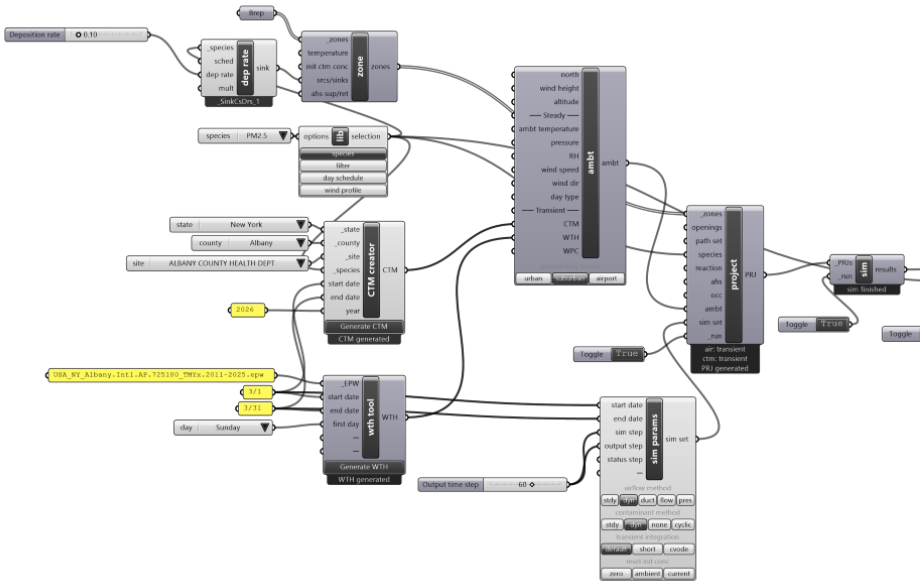


Figure 3 Grasshopper/ANT workflow for transient multizone $PM_{2.5}$ simulation of ETEC.

RESULTS AND DISCUSSION

Period-Level Variation

Table 1 summarizes the observed and simulated period-mean indoor $PM_{2.5}$ concentrations. The observed five-room mean

decreased from 5.70 µg/m^3 (2.49 × 10^{-6} grain/ft^3) during the academic period to 4.72 µg/m^3 (2.06 × 10^{-6} grain/ft^3) during spring break, corresponding to a 17.1% reduction. The model reproduced the direction of this change, with the simulated mean decreasing from 4.52 to 3.87 µg/m^3 (1.97 × 10^{-6} to 1.69 × 10^{-6} grain/ft^3). The weekday-to-weekend reduction was 3.5% during the academic period and 19.0% during spring break, indicating that temporal differences became more pronounced when regular academic activities were reduced.

Table 1. Period-Mean Indoor $PM_{2.5}$ Concentrations and Absolute Percent Bias

Location	Academic period		APB	Break period		APB
	Obs.	Sim.	(%)	Obs.	Sim.	(%)
ETEC B010	5.87	4.36	25.9	4.97	3.75	24.6
Basement Café	9.79	4.74	51.6	8.03	4.06	49.5
ETEC 0149A	5.85	4.25	27.3	4.95	3.64	26.5
ETEC 107	3.53	4.55	29.0	2.88	3.89	35.1
ETEC 103	3.45	4.69	36.0	2.79	4.01	43.4
Indoor mean	**5.70**	**4.52**	**33.9**	**4.72**	**3.87**	**35.8**

Note: Concentrations are µg/m^3; 1 µg/m^3 = 4.369 × 10^{-7} grain/ft^3. APB = absolute percent bias.

Room-Level Model Performance

Room-level behavior was more heterogeneous than the building-level summary. The basement café had the highest observed mean during both the academic and spring-break periods, 9.79 and 8.03 µg/m^3 (4.28 × 10^{-6} and 3.51 × 10^{-6} grain/ft^3), respectively, and was underpredicted by approximately 50%. This persistent underprediction likely reflects cooking emissions, occupant activity, particle resuspension, or local airflow pathways not represented in the current model. ETEC B010 and ETEC 0149A were also underpredicted, whereas ETEC 107 and ETEC 103 were overpredicted.

Figure 4 illustrates the contrasting model behavior in the basement café and ETEC 103 during the academic period. The mean room-specific absolute percent bias was 33.9% for the academic period and 35.8% for spring break. These room-dependent residuals indicate that outdoor infiltration and uniform model assumptions alone cannot explain the observed spatial variability. These discrepancies may also arise partly from temporal variability in actual building operations and space-use patterns, including HVAC schedules, ventilation settings, room use, and occupancy, which can vary with academic activities over time.

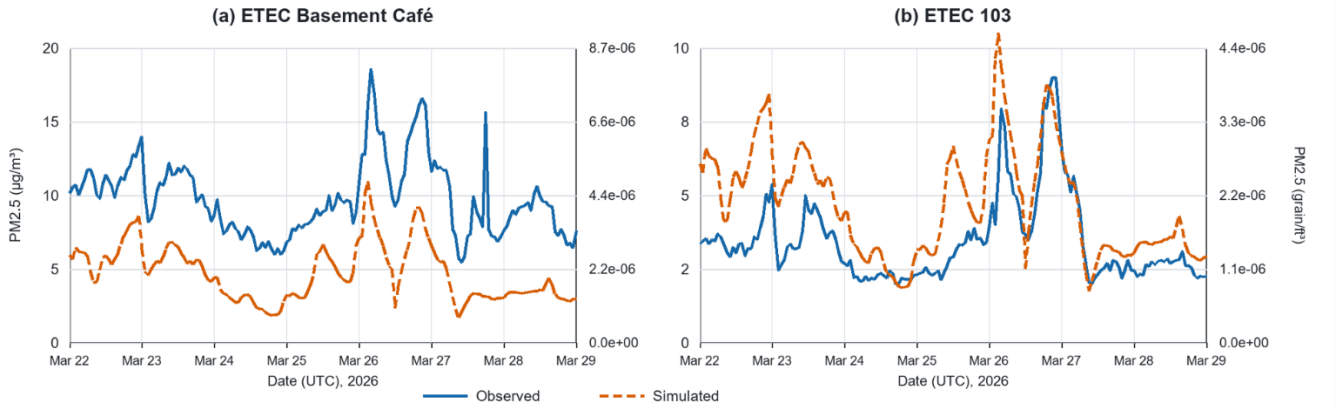


Figure 4 Hourly observed (solid) and simulated (dashed) $PM_{2.5}$ during the academic period for (a) ETEC Basement Café and (b) ETEC 103. Right axes give equivalent I-P units.

Future work will incorporate continuous outdoor monitoring, room-specific HVAC and filtration schedules, door operation, interzonal airflow, and indoor particle sources. CO_2, motion, and light measurements will be evaluated as indicators of occupancy and activity. The monitoring–simulation framework will also be tested across additional seasons, high-smoke periods, and campus buildings.

CONCLUSIONS

Distributed monitoring and multizone simulation revealed clear occupancy-period and room-level differences in indoor

$PM_{2.5}$ within the ETEC building. The observed five-room mean decreased by 17.1% from the academic period to spring break, and the simulation reproduced the direction of this period-to-period change.

Model performance varied substantially among rooms. The basement café had the highest measured concentrations and was underpredicted by approximately 50% during both periods. ETEC B010 and ETEC 0149A were also underpredicted, whereas ETEC 103 and ETEC 107 were overpredicted. The room-specific errors demonstrate the importance of explicitly representing occupancy, indoor particle sources, and local transport pathways when calibrating multizone $PM_{2.5}$ models.

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